

CFD Simulation of a Fully Developed Turbulent Flow in an Annular Pipe

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Abstract

This report presents a simulation of turbulent flow in a circular tube, analyzed in two dimensions. The main goal is to explore how the turbulent flow develops, including examining the velocity profile and key flow characteristics. The simulation investigates how the flow changes over time under the given conditions, offering a deeper understanding of the turbulent flow's behavior and the velocity distribution. The findings provide valuable data to better understand the dynamics of turbulent flow in pipes and guide further analysis.

1 Introduction

When a fluid enters a circular pipe at a uniform velocity, the fluid velocity at the pipe wall becomes zero due to the no-slip condition. As the fluid moves away from the wall, its velocity increases, but the internal friction between the fluid layers, or shear stress, creates a velocity gradient, with slower speeds near the wall and faster speeds at the center of the pipe.

In the central region of the pipe, the flow behaves like inviscid flow, with no significant viscous resistance. To maintain constant flow, the slower velocity near the wall is balanced by faster flow in the core. As the flow moves downstream, the boundary layers near the wall grow and merge, marking the transition from the developing flow to the fully developed flow, where the velocity profile stabilizes and no longer changes along the pipe.

In turbulent flow, shear stress results from both viscous and turbulent effects. The flow is divided into three layers:

- **Viscous Sublayer:** Closest to the wall, dominated by viscous forces, where the velocity profile is linear.

- **Overlap Layer:** An intermediate region where both viscous and turbulent effects are significant.
- **Outer Layer:** Farther from the wall, where turbulence dominates, and viscous effects become negligible.

Understanding these layers and the development of the velocity profile is key to simulating turbulent flow in pipes, which is essential for applications such as fluid transport and heat exchangers.

2 Simulation

2.1 Introduction

In this study, we used ANSYS Fluent to simulate turbulent flow in a circular pipe. The process was carried out in two main stages: (i) Pre-processing and (ii) Post-processing.

During the pre-processing stage, we designed the pipe geometry and generated a computational mesh to define the fluid domain. The mesh is crucial for breaking down the flow field and setting the boundary conditions.

After finalizing the mesh and setup, we used ANSYS Fluent to solve the equations governing turbulent flow. Once the simulation was complete, the post-processing phase involved analyzing the results using Fluent visualization tools, which helped us examine important flow characteristics.

2.2 Pre-processing

Pre-processing is the phase of simulation in which we define the geometry of an object, control volume or control surface, mesh, etc.

2.2.1 Geometry of Pipe

We considered a pipe with a length of 20 m and a diameter of 1 m (Figure 1).

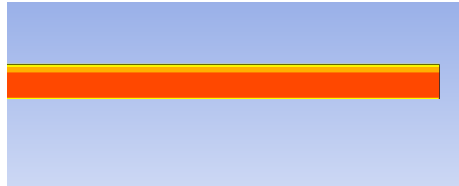


Figure 1: Geometry of the pipe: 20 m in length and 1 m in diameter

2.2.2 Meshing

We divided the wall and the axis into 100 divisions, while the inlet and outlet were divided into 50 divisions each. Additionally, a bias factor was applied to

refine the mesh near the wall, ensuring smaller mesh elements for more accurate calculations and results (Figures 2, 3, and 4).

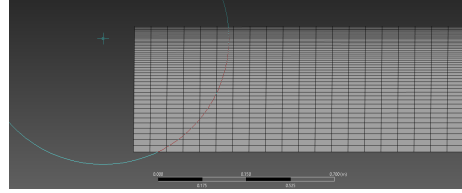


Figure 2: Mesh generation with refinements near the pipe wall

Bias Type	- - - - -
Bias Option	Bias Factor
☒ Bias Factor	10.0
Reverse Bias	1 Edge

Figure 3: Bias type and bias option

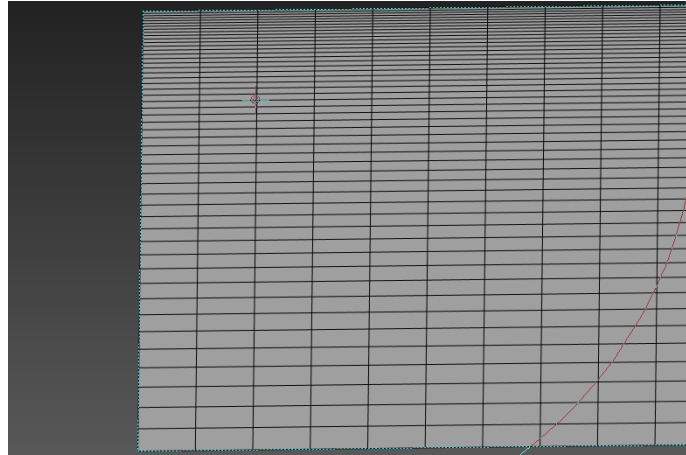


Figure 4: Enhanced mesh accuracy near the wall

2.2.3 Selection of Solver

In this simulation, we used the k-omega SST (Shear Stress Transport) model to capture turbulence more accurately, especially near the pipe wall. This model combines the strengths of the k-omega model in the near-wall region with the k-epsilon model in the free stream, making it ideal for situations where precision is needed close to the wall. The k-omega SST model is particularly effective for

predicting turbulence in complex flows, providing more reliable results in areas with high shear stresses and boundary layer effects.

2.2.4 Boundary Conditions

1. Inlet Boundary:

- **Inlet Velocity:** The fluid enters the pipe with a uniform velocity of 1 m/s.
- **Turbulence Intensity:** Set to 4.8 percent (Figure 5), representing typical turbulence conditions for high Reynolds number flows.
- **Hydraulic Diameter:** The pipe's hydraulic diameter is 1 meter, which defines the geometric properties of the flow domain.

2. Wall Boundary:

- The pipe walls are set as no-slip boundaries, meaning the fluid velocity at the wall is zero.

3. Axis Boundary:

- The pipe axis is treated as an axis of symmetry, meaning no flow or velocity is assumed along this boundary, which helps model the pipe as a 2D flow domain.

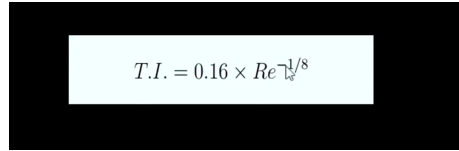

$$T.I. = 0.16 \times Re^{-1/8}$$

Figure 5: Turbulence Intensity at the Inlet

2.3 Post-processing

The solution data were gathered after iterations converged and are represented in Figure 6.

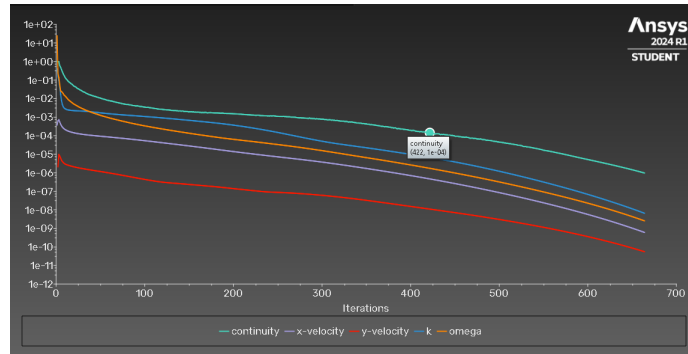


Figure 6: Simulation results after 664 iterations

3 Result and Discussion

The following figures (7, 8, 9, and 10) show the changes in the velocity profile as the flow progresses through the pipe.

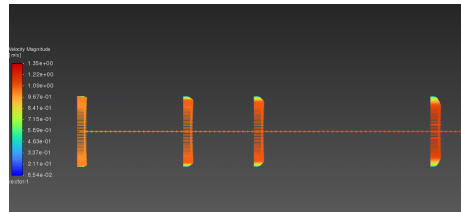


Figure 7: Change in velocity profile along the pipe

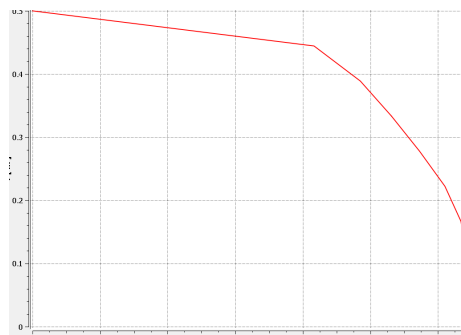


Figure 8: Velocity profile at 15 m downstream

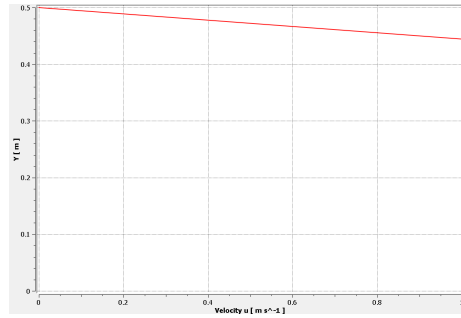


Figure 9: Velocity profile at the inlet

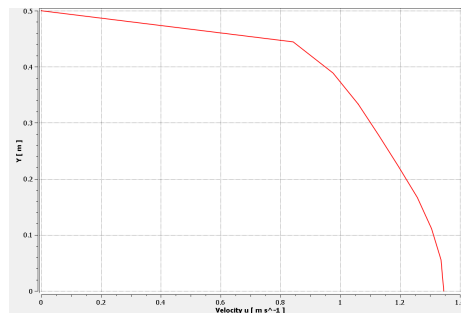


Figure 10: Velocity profile at the outlet

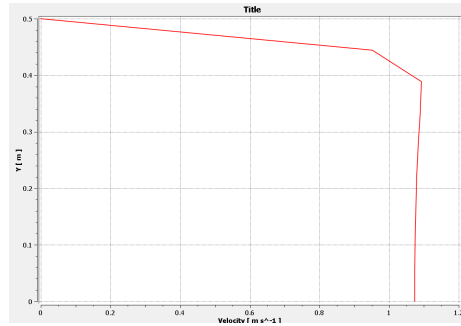


Figure 11: Velocity distribution at 25 m along the Y-axis

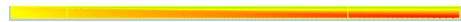


Figure 12: developed flow in pipe 20 m

After a certain distance from the entrance, the boundary layers merge, and the inviscid core flow disappears. The region from the entrance to the point

where the boundary layers merge is known as the **hydrodynamic entrance region**, and the length of this region is referred to as the **hydrodynamic entry length** (L_e). Within this region, the flow is termed **developing flow** because the velocity profile $u(x, r)$ continues to evolve. Downstream of the entrance region, the velocity profile becomes **fully developed** and remains unchanged with respect to the axial position x , resulting in constant wall shear stress. This condition is referred to as **fully developed flow**.

As shown in Figures 7, 8, 9, and 10, the velocity profile initially exhibits some irregularities due to the high Reynolds number, which causes chaotic motion of the fluid particles. A longer distance is required for the flow to develop fully and stabilize, resulting in the entry length still being in the process of developing flow. The theoretical value indicates that the entry length is approximately 22 meters (Figure 16).

$$\frac{Le}{D} = 4.4 * Re_D^{1/6}$$

Figure 13: The entry length

3.1 Another simulation on longer pipe

We conducted an additional simulation using the same strategy outlined in the earlier sections but extended the pipe length to 30 meters. In this simulation, as depicted in Figure 14, the velocity profile becomes stable and fully developed between 25 and 30 meters. This stabilization indicates that the flow has transitioned into the fully developed region, as further illustrated in Figures 15 and 16.



Figure 14: Developing flow



Figure 15: 25 to 30

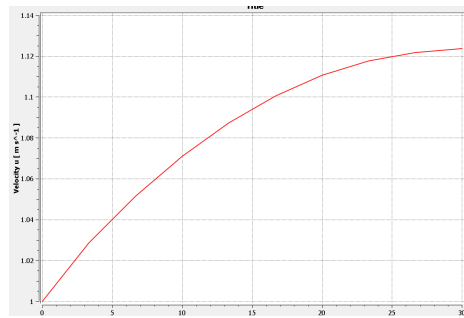


Figure 16: velocity at the axis from the 0 to 30